Introduction to Quantitative Biology
Lecture Notes: Error Rates and Kinetic Proofreading

From An Introduction to Systems Biology — Uri Alon

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The principle of **robustness** states that biological systems have special designs that make their essential function work precisely despite naturally occurring noise. To define robustness, we ask which functions are robust with respect to which source of noise or errors.

In this lecture, the errors result from the presence, for each molecule X, of many chemically similar molecules that can confound its specific recognition. The core question is: how can a biochemical recognition system pick out a specific molecule in a sea of other molecules that bind it with only slightly weaker affinity?

We will see that diverse molecular recognition systems in the cell employ the same principle to achieve high precision. This principle is called **kinetic proofreading**, which was proposed independently by John Hopfield (1974) and Jacques Ninio (1975).

We will discuss different proofreading strategies used by proteins:

- how kinetic proofreading of the genetic code during translation can reduce error rates through irreversible, nonequilibrium reactions;
- how kinetic proofreading based on time delays improves recognition of self and non-self in the immune system;
- and how conformational proofreading provides specificity without consuming energy.

1. Error Rates in Replication, Transcription, and Translation

We begin with a few calculations to build intuition about error rates in DNA replication, transcription, and translation in bacteria and humans. For scale, the human genome (*Homo sapiens*) contains approximately 3 Gbp and 20,000 genes, whereas the genome of *E. coli* contains approximately 5 Mbp and 4,000 genes.

- **Replication: DNA to DNA**

  How many mutations occur in *E. coli* per replication? Take an error rate of $\sim 10^{-10}$ mutations per base per replication. Using $\sim 10^7$ copied bases as an order-of-magnitude estimate for a bacterial replication gives

  $$10^{-10}\ \frac{\text{mut.}}{\text{base} \cdot \text{rep.}} \times 10^7\ \text{bases} = 10^{-3}\ \frac{\text{mut.}}{\text{rep.}}$$

  Thus the mean is about one mutation per thousand genome replications. Each bacterium carries about 0.001 mutations. A large population can nevertheless contain many mutations: 10^{10} such replications give an expected $10^{-3} \times 10^{10} = 10^7$ mutations in total. So if you are looking for a specific mutation, it may already be there.

- **Transcription: DNA to mRNA**

  How many mRNAs have a transcriptional error in them? For an mRNA of 900 nucleotides, assume the error rate is $p = 10^{-4} \sim 10^{-5}$ per nucleotide.

Let L denote the number of nucleotides in an mRNA and p the error probability per nucleotide. Assuming errors at different positions are independent, the probability that an mRNA has no errors is $(1-p)^L$. Using the small-x approximation

$$\ln(1+x) \approx x \qquad (|x| \ll 1),$$

we obtain

$$\ln\left[(1-p)^L\right] = L\ln(1-p) \approx -Lp,$$

so the probability that an mRNA contains at least one error is

$$P(\text{at least one error}) = 1-(1-p)^L \approx 1-e^{-Lp}.$$

For $L = 900$ nucleotides, this gives

$$1-(1-10^{-4})^{900} \approx 0.086$$
$$1-(1-10^{-5})^{900} \approx 0.009$$

Thus roughly 1%–10% of these transcripts contain at least one error.

- **Translation: mRNA to protein**

  How many proteins have a translational error in them? For a protein of 300 amino acids, assume the error rate is $p = 10^{-3} \sim 10^{-4}$ per amino acid. The same calculation applies, now taking L as the number of amino acids and p as the error probability per amino acid.

$$1-(1-10^{-3})^{300} \approx 0.259$$
$$1-(1-10^{-4})^{300} \approx 0.030$$

  Thus roughly 3%–26% of these proteins contain at least one error.

These estimates show why precision at each step matters. To understand how cells achieve this precision, we now focus on translation. The observed translation error rate is about 10^{-4}, whereas the affinity ratio of correct and incorrect tRNAs predicts an equilibrium recognition error of about 10^{-2}. Kinetic proofreading provides a mechanism to explain this discrepancy.

2. Kinetic Proofreading of the Genetic Code Can Reduce Error Rates

In translation, a ribosome produces a protein by linking amino acids one by one into a chain. The type of amino acid added at each step to the elongating chain is determined by the information encoded in an mRNA. Each of the 20 amino acids is encoded by a codon, a series

of three letters on the mRNA. The mapping between the 64 codons and the 20 amino acids is called the **genetic code**.

Each amino acid is brought into the ribosome connected to a specific tRNA molecule. That tRNA has a three-letter recognition site that pairs with the codon sequence for that amino acid on the mRNA. This molecular recognition process works under thermal noise, and thus has an error rate. The wrong tRNA can attach to the codon, resulting in a translation error where a wrong amino acid is incorporated into the elongating protein. These translation errors occur at a frequency of about 10^{-4}. This means that a typical protein of 100 amino acids has a 1% chance to have one wrong amino acid.

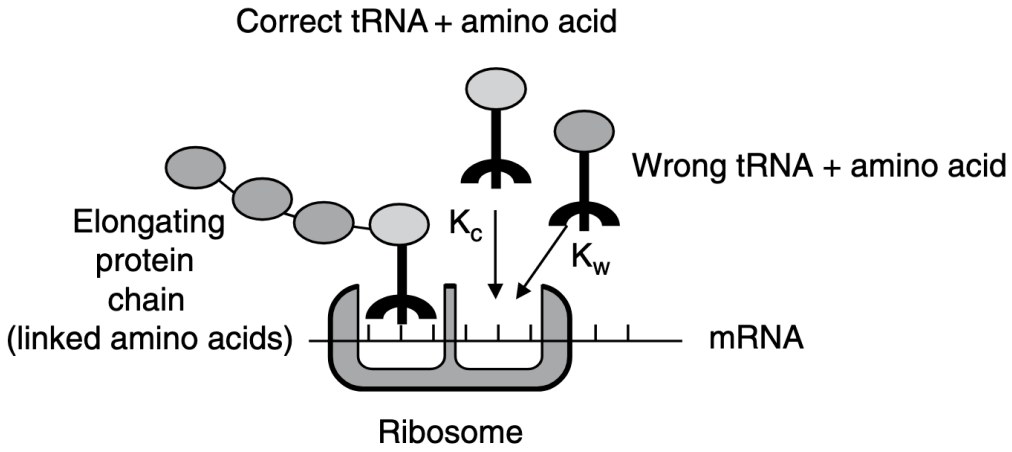


Figure 1: Kinetic Proofreading of the Genetic Code

2.1 Equilibrium Binding Cannot Explain the Precision of Translation

The simplest model for this recognition process is **equilibrium binding** of tRNAs to the codons: forward binding and reverse dissociation balance. Consider codon C on the mRNA in the ribosome. Codon C binds the correct tRNA, denoted c, with an **on-rate** k_c, the association rate constant. The tRNA unbinds from the codon with **off-rate** k_c', the probability of dissociation per unit time. When the tRNA is bound, there is a probability v per unit time that its amino acid will be covalently linked to the growing protein chain:

$$c + C \underset{k_c'}{\overset{k_c}{\rightleftharpoons}} [cC] \xrightarrow{v} \text{correct amino acid}$$

At equilibrium, the concentration of the complex $[cC]$ is given by the balance of binding and dissociation. Neglecting the slow linking reaction, $k_c cC = k_c'[cC]$. Hence,

$$[cC] = \frac{cC}{K_c}$$

where the **dissociation constant** is equal to the ratio of the off-rate and on-rate:

$$K_c = \frac{k'_c}{k_c}$$

The smaller the dissociation constant, the higher the affinity of the reactants. The incorporation rate of the correct amino acid is

$$R_{\text{correct}} = v[cC] = \frac{vcC}{K_c}$$

Here c and C also denote free concentrations; $[cC]$ denotes the concentration of the bound complex.

In addition to the correct tRNA, the cells contain different tRNAs that carry the other amino acids. These tRNAs compete for binding to codon C. Let us consider only the most dangerous incorrect tRNA, denoted w: the tRNA that carries a different amino acid and has the highest affinity to codon C. Its concentration is about equal to the concentration of the correct tRNA. The corresponding process is

$$w + C \underset{k'_w}{\overset{k_w}{\rightleftharpoons}} [wC] \xrightarrow{v} \text{wrong amino acid}$$

with wrong amino acid incorporation rate

$$R_{\text{wrong}} = \frac{vwC}{K_w}$$

The linkage rate v is the same for the correct and wrong tRNAs because linkage occurs at a distant site on the ribosome, away from the site of codon recognition.

The resulting **error rate**, F_o, is the ratio of the rates of incorrect and correct amino acid incorporation. Since the tRNA concentrations are about the same,

$$F_o = \frac{R_{\text{wrong}}}{R_{\text{correct}}} = \frac{vwCK_c}{vcCK_w} \approx \frac{K_c}{K_w}$$

The on-rates for both w and c are limited by diffusion and are about the same, $k_w = k_c$. It is the off-rate which distinguishes the correct codon from the incorrect one: the wrong tRNA unbinds more rapidly, $k'_w \gg k'_c$. Using the off-rate ratio,

$$F_o = \frac{R_{\text{wrong}}}{R_{\text{correct}}} = \frac{K_c}{K_w} \approx \frac{k'_c}{k'_w}$$

The affinity ratio is about $K_c/K_w \sim 1/100$. There is a large discrepancy between the predicted equilibrium recognition error, $F_o \sim K_c/K_w \sim 1/100$, and the actual translation error rate, $F = 1/10{,}000$. Equilibrium recognition cannot explain the high fidelity found in this system.

2.2 Kinetic Proofreading Can Dramatically Reduce the Error Rate

What mechanism can explain the high fidelity of the translation machinery, which is 100-fold higher than predicted from equilibrium recognition? The solution lies in a reaction that occurs in the translation process. After binding the codon, c is converted to c^*. This reaction is virtually irreversible, because it is coupled to the hydrolysis of a GTP molecule. The modified state can either fall off the codon or donate its amino acid to the elongating protein chain:

$$\begin{array}{ccccccc} c + C & \underset{k_c'}{\overset{k_c}{\rightleftharpoons}} & [cC] & \xrightarrow{m} & [c^*C] & \xrightarrow{v} & \text{correct amino acid linked} \\ & & & & \downarrow l_c' & & \\ & & & & c^* + C & & \end{array}$$

This process costs energy, however, it is precisely this design that generates high fidelity. The secret is that c^* offers a second discrimination step: the wrong tRNA, once modified, can fall off the codon, but it cannot mount back on.

The concentration of the modified bound complex is given by the balance of its formation and dissociation, since v is much smaller than the other rates:

$$m[cC] = l_c'[c^*C], \qquad [c^*C] = \frac{m[cC]}{l_c'}$$

The rate of correct incorporation is

$$R_{\text{correct}} = v[c^*C] = \frac{vmcC}{l_c' K_c}$$

The same applies for the wrong codon w. The conversion of w to w^* occurs at the same rate m, because the modification process does not discriminate between tRNAs. The off-rate ratio of the correct and incorrect modified tRNAs is the same as the ratio for the unmodified tRNAs:

$$\frac{l_w'}{l_c'} = \frac{k_w'}{k_c'} \approx \frac{K_w}{K_c}$$

The rate of wrong amino acid linkage is

$$R_{\text{wrong}} = v[w^*C] = \frac{vmwC}{l_w' K_w}$$

resulting in an error rate

$$F = \frac{R_{\text{wrong}}}{R_{\text{correct}}} = \left(\frac{K_c}{K_w}\right)\left(\frac{l_c'}{l_w'}\right) = \left(\frac{K_c}{K_w}\right)^2$$

The irreversible reaction step affords a proofreading event that adds a multiplicative factor of K_c/K_w to the error rate. It effectively works like two separate recognition processes, the second working on the output of the first:

$$F = F_o^2$$

The proofreading step can reduce the error rate from $F_o = 1/100$ to $F = F_o^2 = 1/10{,}000$.

A two-stage fishing analogy. Imagine a river containing equal numbers of golden and black fish, where only golden fish are desired. A selective bait catches each golden fish with probability $q_g = 0.9$, but each black fish with probability $q_b = 0.009$. Starting with 10^6 fish of each color, the first round catches 900,000 golden fish and 9,000 black fish. The fraction of black fish among those caught is

$$P(\text{black} \mid \text{passed round 1}) = \frac{9{,}000}{900{,}000 + 9{,}000} \approx 10^{-2}.$$

Now place only the fish caught in the first round in a separate pond and subject these same fish to the same selective bait a second time. Assuming independent outcomes in the two rounds conditional on fish color, the expected numbers passing both rounds are $900{,}000 \times 0.9 = 810{,}000$ golden fish and $9{,}000 \times 0.009 = 81$ black fish. Thus,

$$P(\text{black} \mid \text{passed both rounds}) = \frac{81}{810{,}000 + 81} \approx 10^{-4}.$$

Rejected fish are simply absent from the final catch. The aim is not to prevent wrong fish from being caught initially, but to reduce the **conditional probability of a wrong final product**. The second round tests only fish that passed the first; it is not an independent catch from the original river. Likewise, in kinetic proofreading a wrong tRNA may bind initially, but dissociation at either discrimination step prevents its incorporation into the final protein.

Had all reactions been reversible and at equilibrium, no improvement would be gained over the simple scheme. This is due to detailed balance: at equilibrium, the forward and reverse fluxes of each elementary transition balance.

An even higher level of fidelity can be attained by linking together n irreversible proofreading processes. Each irreversible step adds a proofreading factor F_o, resulting in an overall error rate of

$$F = F_o^{n+1}$$

DNA damage recognition by UvrABC. In bacterial nucleotide excision repair, UvrA first recognizes a candidate DNA lesion and recruits UvrB for additional damage verification. ATP-dependent transitions involving UvrA and UvrB provide opportunities for incorrect complexes to dissociate before UvrC makes incisions around the lesion. These successive checks provide another example of kinetic proofreading in molecular recognition. [6]

3. Recognition of Self and Non-Self by the Immune System

The immune system uses a different way of kinetic proofreading, based on time delays.

One of the important roles of the immune system is to scan the cells of the body for antigens, such as fragments of proteins made by a virus that has infected the cell. The scanning task is carried out by T-cells. Each of the T-cells has receptors that are specific against a foreign protein antigen. Cells present fragments of proteins in dedicated protein complexes on the cell surface called MHCs. If the T-cell receptor recognizes its antigen in an MHC, it triggers a signal-transduction cascade inside the T-cell.

In the recognition process, it is essential that the T-cell does not kill cells that present proteins that are normally produced by the healthy body. If such misrecognition occurs, the immune system attacks the cells of the body, potentially leading to an autoimmune disease.

The precision of the recognition of non-self-proteins by T-cells is remarkable. T-cells can recognize minute amounts of a foreign protein antigen in a background of self-proteins, even though the self-proteins have only a slightly lower affinity to the T-cell receptor than the foreign target. The error rate of recognition is less than 10^{-6}, although the affinity of the antigen is often only 10-fold higher than the affinities of the self-proteins.

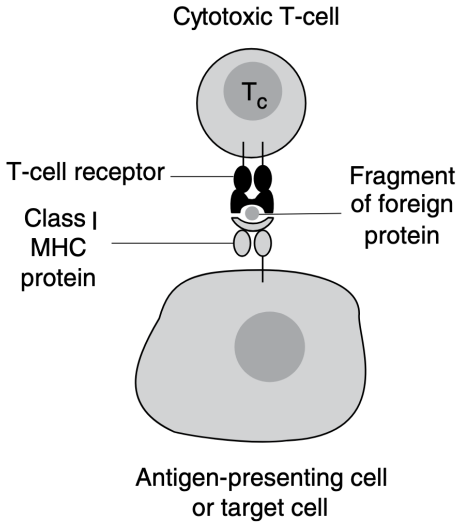


Figure 2: The Immune System

3.1 Equilibrium Binding Cannot Explain the Low Error Rate of Immune Recognition

The receptors on a given T-cell recognize a specific foreign protein, which we call the correct ligand c. In addition to c, the receptors are exposed to a variety of self-proteins with weaker affinity. For clarity, we treat these wrong ligands as a single entity w.

The correct ligand binds receptor R at rate k_{on} and dissociates with rate k_{off}, with $K_c = k_{\text{off}}/k_{\text{on}}$. Once ligand binds the receptor, the signaling pathway is activated with probability v per unit time. Therefore,

$$A_{\text{correct}} = v[cR] = \frac{vcR}{K_c}$$
$$A_{\text{wrong}} = v[wR] = \frac{vwR}{K_w}$$

The error rate of the T-cells, defined by the ratio of incorrect to correct activations, is

$$F_o = \frac{A_{\text{wrong}}}{A_{\text{correct}}} = \left(\frac{K_c}{K_w}\right)\left(\frac{w}{c}\right)$$

The incorrect ligands often have only a 10-fold lower affinity, $K_c/K_w \sim 0.1$. Their concentration often greatly exceeds that of the pathogen protein. Hence $F_o > 0.1$, far higher than the observed recognition error rate in T-cell recognition, 10^{-6} or lower.

3.2 Kinetic Proofreading Increases Fidelity of T-Cell Recognition

The actual recognition process in T-cell receptors includes several additional steps. After ligand binding, the receptor undergoes a series of covalent modifications, such as phosphorylation on numerous sites. These modifications are energy-consuming and are held away from thermal equilibrium. Activation begins only after these modifications and binding events are complete.

The extra steps create a **delay** τ that allows the system to reduce its error rate: only ligands that remain bound to the receptors for a long enough time have a chance to activate the T-cell. Experimentally, the cutoff dwell time for activation is about 3–5 seconds.

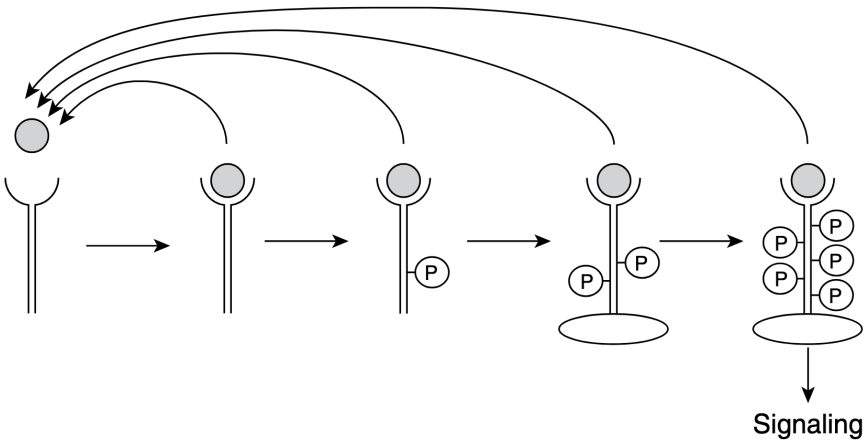


Figure 3: Kinetic Proofreading of T-Cell Recognition

Once bound, the correct ligand has a probability per unit time k_{off} to dissociate. The probability that it remains bound for longer than time t is

$$P(t) = e^{-k_{\text{off}}t}$$

Signaling in the cell only occurs at a delay τ after the ligand binds the receptor, due to the series of modifications of the receptors needed to activate the signaling pathway. Hence, the probability per ligand binding event that the T-cell is activated is

$$A_{\text{correct}} = e^{-k_{\text{off}}\tau}$$

The incorrect ligand has a larger off-rate k'_{off}, so the probability that the incorrect ligand activates the receptor is

$$A_{\text{wrong}} = e^{-k'_{\text{off}}\tau}$$

The error rate in the delay mechanism is the ratio of these activation rates,

$$F = \frac{A_{\text{wrong}}}{A_{\text{correct}}} = e^{-(k'_{\text{off}}-k_{\text{off}})\tau}$$

This allows a very small error rate even for moderate differences between the off-rates, provided that the delay is long enough. For example, if $k_{\text{off}} = 0.5\ \text{s}^{-1}$, $k'_{\text{off}} = 5\ \text{s}^{-1}$, and $\tau = 3$ s,

$$F \sim e^{-(5-0.5)\cdot 3} = e^{-13.5} \sim 10^{-6}$$

Thus, long delays can enhance **fidelity**. However, the longer the delay, the larger the number of binding events of the correct ligand that unbind before signaling can begin. Increasing the delay can cause a loss of **sensitivity**, the ability to respond to the correct ligand.

The hallmark of kinetic proofreading is a nonequilibrium reaction in the recognition process that forms an intermediate state, providing a delay after ligand binding. These ingredients are found in diverse recognition processes in the cell.

4. Conformational Proofreading Provides Specificity without Consuming Energy

Kinetic proofreading depends on energy-burning steps, such as ATP hydrolysis. A different mechanism for specificity works at equilibrium without burning ATP.

Proteins are flexible, and they deform when they bind the ligand—a phenomenon called **induced fit**. Induced fit was puzzling, because it costs elastic energy to deform the proteins, and thus reduces binding of the ligand. But most proteins show induced fit. Savir and Tlusty [3, 4] provided a fresh view by showing that induced fit is a good idea if the goal is not to bind tightly but to avoid binding to the wrong partner. This mechanism is called **conformational proofreading**.

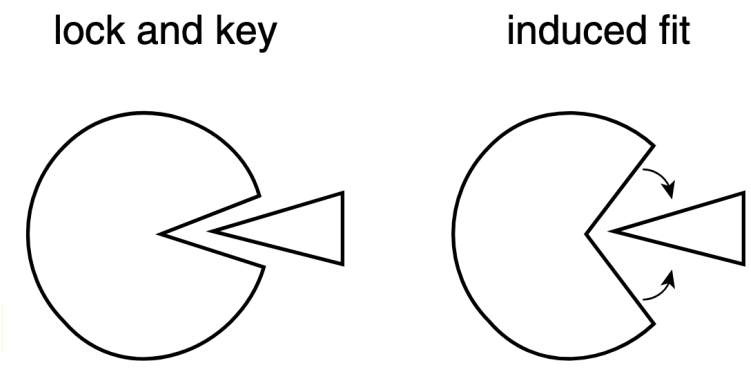


Figure 4: Induced fit

In conformational proofreading, one adds an energy handicap h to the free energy of binding. The elastic bending energy of induced fit is such a handicap. The handicap gets added to the free energies of binding of the correct and wrong targets, ΔG_c and ΔG_w, which become $\Delta G_c + h$ and $\Delta G_w + h$.

Since the binding probability varies as $1/(1 + e^{\Delta G})$, the handicap affects the wrong target much more strongly than the correct target. There is an optimal bending energy h at which the binding of the correct target is reduced slightly, but the binding of the wrong target is much reduced. (Figure 5)

More explicitly, with binding free energies expressed relative to the unbound state, the probability of binding target i is

$$P_i(h) = \frac{1}{1 + e^{(\Delta G_i + h)/k_B T}}, \qquad i \in \{c, w\}.$$

Even when $h = 0$, the two binding probabilities are not identical: because the correct target has a lower binding free energy ($\Delta G_c < \Delta G_w$), it has a higher probability of binding ($P_c(0) > P_w(0)$). When both free energies are sufficiently negative, however, both probabilities are close to one and the targets are difficult to distinguish. Increasing h raises both free energies by the same amount, so their difference $\Delta G_w - \Delta G_c$ remains unchanged. Nevertheless, the **nonlinear** dependence of binding probability on free energy allows the wrong-target probability to fall substantially while the correct-target probability remains high.

For example, take $\Delta G_c = -6k_BT$ and $\Delta G_w = -2k_BT$. Without bending ($h = 0$), the probabilities are $P_c \approx 0.998$ and $P_w \approx 0.881$. With $h = 4k_BT$, they become $P_c \approx 0.881$ and $P_w \approx 0.119$. The difference in binding probabilities increases from about 0.117 to 0.762, even though the difference in binding free energies remains $4k_BT$. This is the principle of conformational proofreading.

Conformational proofreading predicts the optimal degree of bending that maximizes specificity. This agrees with the sizable bend that ribosomes make when binding tRNA. [5]

Conformational proofreading also helps explain how the homologous-recombination protein RecA searches for matching DNA sequences in a large genome. In *E. coli*, RecA forms a nucleoprotein filament on single-stranded DNA and extends the bound DNA by about 50% relative to B-form DNA. The filament then probes candidate double-stranded DNA molecules

for homology. This deformation introduces an energetic and geometric constraint that can favor recognition of homologous over mismatched sequences. A theoretical analysis found that the observed RecA–DNA binding energetics are close to optimal for sequence detection and proposed a specificity-enhancing role for this DNA extension. [4]

In ribosomes and RecA, conformational and kinetic proofreading work together to enhance specificity.

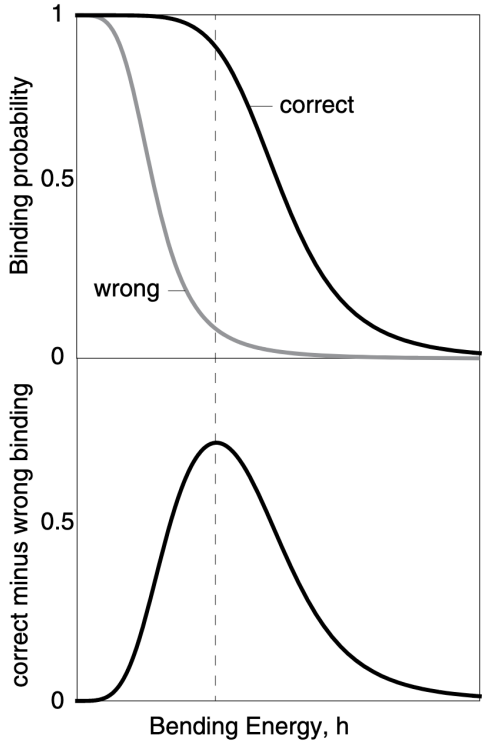


Figure 5: Conformational Proofreading

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